

Cross-Lingual Hallucination Drift in LLMs: Does It Depend on Task Type?

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Abstract

Large Language Models (LLMs) are known to hallucinate generating factually incorrect content with high confidence and this problem is widely assumed to worsen in non-English languages due to training data imbalances. A recent large-scale study by [ul Islam et al. \(2025\)](#) found a surprising result: *length-normalized hallucination rates are uncorrelated with language resource level*, overturning the intuitive assumption that low-resource languages hallucinate more. However, that study evaluated hallucination exclusively on open-domain, knowledge-intensive long-form QA using open-source LLMs. It is an open question whether this finding holds across *different task types*. We hypothesize that the relationship between language resource level and hallucination rate is **task-dependent**: drift may be significant for factual recall tasks but negligible for commonsense reasoning tasks, or vice versa. This empirical project tests that hypothesis by measuring hallucination rates of GPT-4o-mini across three languages (English, Spanish, Swahili) on two structurally different tasks, factual QA (TruthfulQA) and commonsense reasoning (XCOPA) using an LLM-as-a-Judge evaluation pipeline. Our key research question is: *does cross-lingual hallucination drift vary by task type, and does the resource-level correlation depend on the nature of the task?*

1 Introduction

Hallucination in LLMs where a model generates non-factual or unfaithful content, often with a confident tone is a well-documented failure mode in English-language settings ([Alansari et al., 2025](#)). As LLMs are deployed globally, a natural concern is whether hallucination rates are worse in non-English languages, particularly low-resource ones where training data is sparse ([Blasi et al., 2022](#)). This assumption has motivated significant interest in *Cross-Lingual Hallucination Drift*: the change

in hallucination rate when a model operates in a non-English language.

A pivotal recent study, [ul Islam et al. \(2025\)](#), conducted the most comprehensive investigation to date, measuring token-level hallucination rates across 30 languages and 6 open-source LLM families on open-domain knowledge-intensive QA. Their surprising finding: length-normalized hallucination rates are *uncorrelated* with language resource level. Low-resource languages like Sindhi (5.8% HR) had *lower* hallucination rates than high-resource languages like Hebrew (16.8% HR). This directly challenges the common assumption underlying much prior work, including our own initial framing.

However, [ul Islam et al.](#)’s study has an important limitation: it evaluates a single task type open-domain long-form QA using Wikipedia as a knowledge source. It is entirely possible that the resource-level correlation of hallucination rates is **task-dependent**. Commonsense reasoning (e.g., XCOPA) relies on world knowledge that may be more culturally and linguistically grounded, while factual recall (e.g., TruthfulQA) targets model-internal parametric knowledge ([Qi et al., 2023](#)). These two task types may expose fundamentally different failure modes across languages.

Our project investigates this gap with the following research question:

Does cross-lingual hallucination drift vary by task type? Does the (lack of) correlation between hallucination rate and language resource level hold for both factual QA and commonsense reasoning?

We measure hallucination rates for GPT-4o-mini a closed, instruction-tuned model not studied by [ul Islam et al.](#) across English (high-resource), Spanish (mid-resource), and Swahili (low-resource), on both TruthfulQA and XCOPA. By comparing drift

patterns across these two task types, we aim to determine whether task type is a confounding variable in cross-lingual hallucination estimation.

We formally define the token-level Hallucination Rate for language l and task t , adopting the precision-recall corrected estimator from [ul Islam et al. \(2025\)](#):

$$HR_{est,l,t} = \frac{P_{l,t} \cdot H_{det,l,t}}{R_{l,t} \cdot N_{l,t}} \times 100 (\%) \quad (1)$$

where $P_{l,t}$ and $R_{l,t}$ are precision and recall of our hallucination detector for language l on task t , $H_{det,l,t}$ is detected hallucination tokens, and $N_{l,t}$ is total tokens generated. Normalizing by $N_{l,t}$ is critical: [ul Islam et al.](#) showed that LLMs produce longer responses in high-resource languages, inflating raw counts without increasing the true rate. We extend their formula by conditioning on task type t , enabling direct comparison of HR across tasks within the same language.

We then define the **Cross-Lingual Drift** for task t as:

$$\Delta HR_{l,t} = HR_{est,l,t} - HR_{est,English,t}$$

If $\Delta HR_{l,t}$ is large for factual QA but near zero for commonsense reasoning (or vice versa), this constitutes evidence that hallucination drift is task-dependent a novel finding that extends and refines the conclusions of [ul Islam et al.](#).

2 Related Work

2.1 Hallucination Benchmarks and Evaluation Metrics

Hallucination evaluation in LLMs has historically been dominated by English-language benchmarks. TruthfulQA ([Lin et al., 2022](#)) introduced 817 questions across 38 domains specifically designed to catch models repeating false beliefs with high confidence, the most widely cited hallucination benchmark and the direct source of our factual QA evaluation dataset. HaluEval ([Li et al., 2023](#)) extended this to 35K human-annotated examples across QA, dialogue, and summarization using binary token-level labeling, establishing the span-level F1 paradigm that our own secondary metric follows. More recently, HalluLens ([Bang et al., 2025](#)) introduced a careful taxonomy distinguishing intrinsic hallucination, extrinsic hallucination, and factuality clarifying that our $HR_{est,l,t}$ specifically targets extrinsic hallucination (outputs inconsistent with the model’s training knowledge), and

raising the data leakage concern we acknowledge in our evaluation design.

MultiHal ([Lavrionics et al., 2025](#)) directly addresses the English-only gap by extending seven established benchmarks (including TruthfulQA and HaluEval) with multilingual and knowledge-graph-grounded coverage, demonstrating that even the most established hallucination benchmarks have been English-centric. [Alansari et al. \(2025\)](#) provide a broad survey of hallucination causes, detection taxonomies, and evaluation metrics across both English and multilingual settings, confirming that cross-lingual hallucination evaluation remains an underexplored area.

2.2 Cross-Lingual Hallucination

The most directly relevant prior work is [ul Islam et al. \(2025\)](#), whose precision-recall-corrected length-normalized HR estimator we adopt as Equation 1. Their key finding that HR is *uncorrelated* with language resource level across 30 languages, is both the primary motivation and the primary challenge for our work. We extend their analysis by decomposing HR by task type, testing whether this null correlation holds separately for factual QA and commonsense reasoning.

[Zhang et al. \(2025\)](#) validate the premise that cross-lingual hallucination is a distinct measurable phenomenon, separate from both cross-modal hallucination and general performance degradation. Their per-language F1 results provide an external comparison point for our own span-level F1 outputs. [Kang et al. \(2024\)](#) evaluate hallucination detection metrics across 19 languages, finding that lexical metrics (ROUGE) are largely ineffective multilingually while NLI-based methods work for high-resource but not low-resource languages. This motivates our use of an LLM-as-a-Judge approach rather than surface-level similarity metrics.

Crucially, [Qi et al. \(2023\)](#) show that even as model scale improves factual accuracy in English, cross-lingual consistency of factual knowledge does not improve at the same rate, the same model may answer a question correctly in English but incorrectly in another language. This is the key empirical grounding for our $\Delta HR_{l,t}$ score: scale does not eliminate drift, and drift may manifest differently depending on whether the question demands parametric recall (TruthfulQA) or commonsense reasoning (XCOPA).

2.3 Language Resource Disparity and LLM Performance

Blasi et al. (2022) established that performance gaps across languages in NLP are systematic and tied to data availability, not random making the investigation of hallucination drift a direct and natural extension of this line of work. Language Ranker (Li et al., 2025) quantifies per-language LLM performance using internal representations and finds strong correlation with pre-training corpus size, reporting that approximately 92.65% of GPT-3’s training tokens are in English. If our $\Delta HR_{l,t}$ scores correlate with Language Ranker scores, that would itself be a meaningful secondary finding. Qin et al. (2024) provide a comprehensive survey of multilingual LLM development covering data imbalances and cross-lingual transfer, establishing the broader landscape of multilingual LLM challenges within which our hallucination-specific analysis sits.

3 Project Outline

3.1 Datasets

We use two datasets representing structurally different task types, allowing us to test whether hallucination drift patterns differ across task categories:

- **Multilingual TruthfulQA** (Lin et al., 2022) : questions designed to probe factual misconceptions that models answer incorrectly with high confidence. This targets *parametric factual recall*: the model must retrieve internally stored facts. Available in English, Spanish, and other languages.
- **XCOPA** (Cross-lingual Choice of Plausible Alternatives) : a commonsense causal reasoning benchmark spanning 11 typologically diverse languages, including Swahili. This targets *commonsense reasoning*: the model must apply world knowledge to select a plausible cause or effect. Its multiple-choice format provides a cleaner signal for hallucination labeling.

3.2 Models

Target model We evaluate **GPT-4o-mini** (OpenAI), queried via API using zero-shot prompts in each target language. ul Islam et al. evaluated only open-source models (Llama, Qwen, Mistral, Gemma, Aya, EuroLLM); closed-model drift behavior remains an open question. Using a single

model controls for model-level confounds when comparing drift across task types.

Judge model We use **GPT-4o** as our LLM-as-a-Judge evaluator (Kang et al., 2024), distinct from ul Islam et al.’s fine-tuned multilingual detector. The judge labels each response as *Hallucinated* or *Faithful* with a brief justification. GPT-4o is selected for its strong multilingual comprehension across all three target languages.

3.3 Software & Tools

All experiments use Python 3.10+ with pandas, scipy, matplotlib, seaborn, the Hugging Face datasets library, and the OpenAI Python SDK. Outputs are stored in structured JSON and documented in Jupyter notebooks.

3.4 Method Pipeline

1. **Data preparation.** Sample 150–200 questions per language per task (TruthfulQA and XCOPA), yielding ~900–1200 total evaluation instances.
2. **Target model inference.** Query GPT-4o-mini with each question in English, Spanish, and Swahili using zero-shot prompting. Log all responses with metadata (language, task, question ID, response text, token count).
3. **LLM-as-a-Judge evaluation.** Submit each (question, response) pair to GPT-4o with the prompt: “Is this response factually accurate? Answer: Hallucinated or Faithful. Briefly justify.” Parse binary labels.
4. **Comparative analysis.** Compute $HR_{est,l,t}$ and $\Delta HR_{l,t}$ for each language–task cell. Produce a 2×3 results table (2 tasks $\times$ 3 languages) and test whether drift patterns differ across tasks using interaction statistics.

4 Experimental Design

4.1 Evaluation Metrics

Primary: Per-task Hallucination Rate. We compute $HR_{est,l,t}$ (Equation 1) separately for TruthfulQA and XCOPA. The key comparison is whether the *pattern* of cross-lingual differences differs across tasks.

Secondary: Task-type drift interaction. We define the **drift interaction score**:

$$\Phi_l = \Delta HR_{l, \text{TruthfulQA}} - \Delta HR_{l, \text{XCOPA}}$$

A large $|\Phi_l|$ indicates hallucination drift is task-dependent for language l . A Φ_l near zero suggests drift is task-agnostic.

Supporting: Span-level F1. Following Li et al. (2023), we spot-check 50 samples per language per task with human annotation and compute span-level F1 to validate judge reliability before trusting aggregate rate estimates.

4.2 Baselines

Baseline 1 — English self-rate per task. GPT-4o-mini evaluated on English TruthfulQA and XCOPA provides task-specific anchors for $\Delta HR_{l,t}$.

Baseline 2 — Translated ground-truth. Machine-translate correct English reference answers (using NLLB) and run the judge on translations. Hallucinations here reflect translation noise, not model drift. Computed separately per task.

Baseline 3 — Dual-judge agreement. Run Claude (claude-sonnet) as a second judge on a 50-sample subset per language per task. Compute Cohen’s κ per cell to verify judge reliability across both task types and languages.

4.3 Success Criteria

Primary: Task-dependent drift. Hypothesis confirmed if $|\Phi_l|$ is statistically significant ($p < 0.05$) for at least one $l \in \{\text{Spanish, Swahili}\}$.

Secondary: Replication of ul Islam et al. Verify whether overall HR is uncorrelated with resource level when aggregated across tasks. Discrepancies are investigated by model type (closed vs. open-source) and evaluation method (LLM-as-a-Judge vs. fine-tuned detector).

Minimum effect size. $|\Phi_l| \geq 2$ percentage points, consistent with the 2–5pp range reported by ul Islam et al..

Reproducibility. Three random seeds; mean $\pm$ std reported. Conditions where std exceeds mean drift are flagged as unreliable.

5 Team Task Distribution

Table 1 outlines the division of responsibilities across the four team members.

Member	Role	Responsibilities
Chun Hsu	Data & Infra	Dataset download and preprocessing; API key setup; data pipeline and JSON storage
Devin Chi	Experiments	Zero-shot inference across 3 languages; output logging; English baseline runs
Bhoomika Monthy Rajashekar	Evaluation	LLM-as-a-Judge prompt design; binary label parsing; span-level F1 computation
Anagha P Krishna	Analysis-Writing	Φ_l and ΔHR computation; statistical tests; visualizations; paper write-up

Table 1: Team task distribution.

6 Proposed Timeline

Table 2 shows our planned schedule from proposal submission through final report.

Week	Milestone
Week 1 (Apr 1–7)	Dataset sampling, API setup, translated ground-truth baseline (NLLB)
Week 2 (Apr 8–14)	Zero-shot inference for all language–task cells, output storage
Week 3 (Apr 15–21)	LLM-as-a-Judge evaluation, dual-judge agreement check, label validation
Week 4 (Apr 22–28)	Compute $HR_{est,l,t}$, $\Delta HR_{l,t}$, Φ_l , statistical tests, visualizations
Week 5 (Apr 29–May 5)	Draft report, internal peer review, revisions and submission

Table 2: Proposed project timeline.

Limitations

Our study covers only three languages; findings may not generalize to languages with non-Latin scripts or more extreme resource imbalance than Swahili (Blasi et al., 2022). Our judge (GPT-4o) is a closed model whose multilingual reliability on Swahili is unknown; we address this through dual-judge agreement but cannot fully rule out judge-language bias (Kang et al., 2024). Zero-shot prompting exclusively may underestimate model capability in non-English settings; few-shot and native-language prompting are left for future work. Finally, our sample of 150–200 questions per condition may produce high-variance estimates, especially for XCOPA’s constrained label space.

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